ELEC5206 Laboratory 1 Report
Safety, Laboratory Introduction and Measurement Equipment

Bench	4	Date	12 August 2026
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1. Aim and equipment

The laboratory aimed to operate the Rigol DG1062Z function generator, Tektronix MDO oscilloscope and Lab-Volt DACI/LVDAC-EMS; verify sine, phase, MATH and PWM measurements; and determine the 82 Ω divider resistance and AC differential voltage from measured data.

Equipment: Rigol DG1062Z; Tektronix MDO 3-series oscilloscope; Lab-Volt 9063 DACI with LVDAC-EMS; 82 Ω/10 W and 40 Ω/10 W resistors; DC supply and leads. Power was removed before wiring changes.

2. Methodology and procedure

Step	Procedure
1	With power off, connect the generator and oscilloscope grounds to circuit ground; verify probe attenuation and channel scales.
2	Set CH1 to a 100 Hz sine wave, 2.00 Vpp, 0° phase and +1.00 V offset; record amplitude and frequency.
3	Set CH2 to 100 Hz, 1.00 Vpp, 0 V offset and +90°; align channels, then measure CH2 relative to CH1.
4	Display CH1 + CH2 with MATH and compare its amplitude and phase with the phasor prediction.
5	Generate a 0-5 V, 100 kHz PWM signal and record frequency and positive/low-state duty at 20%, 40% and 60%.
6	For the DC divider (R_1 = 82 Ω, R_2 = 40 Ω), reduce the supply from 20 V to 10 V in 2 V steps; record E_1, I_1 and P_1 in LVDAC-EMS.
7	For the 50 Hz AC divider, measure VBD on CH1 and VCD on CH2, use CH1 - CH2 for VBC, and compare scope RMS with Lab-Volt RMS.

Data handling: numerical results below are transcribed from the original screenshots/tables. Percentage differences use (measured - reference)/reference × 100%.

3. Function generator and oscilloscope results

The sine-wave settings, two-channel phase relation, CH1 + CH2 MATH trace and 0-5 V PWM output were verified directly on the oscilloscope.

Test	Set value	Measured value	Difference
CH1 amplitude	2.000 Vpp	2.040 Vpp	+2.00%
CH1 frequency	100.000 Hz	100.040 Hz	+0.040%
CH2 phase relative to CH1	+90.000°	+89.892°	-0.108°
CH1 + CH2 AC sum	1.118 Vpk; +26.565°	≈1.1 Vpk at 100 Hz	Consistent

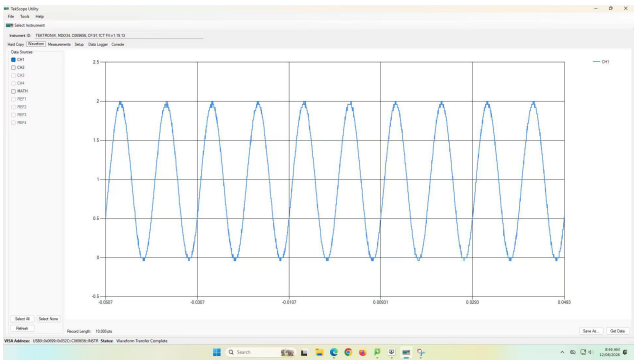


Figure 1. CH1 sine wave (approximately 0-2 V at 100 Hz).

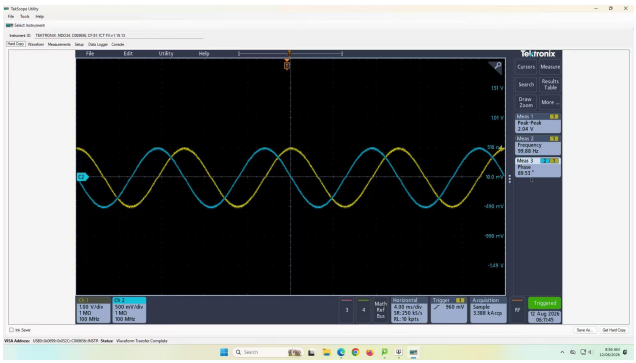


Figure 2. CH1 and CH2 separated by approximately 90°.

$$v_{\mathrm{sum,AC}}(t) = \sin(\omega t) + 0.50\cos(\omega t) = 1.118\sin(\omega t + 26.565°)\ \mathrm{V}$$

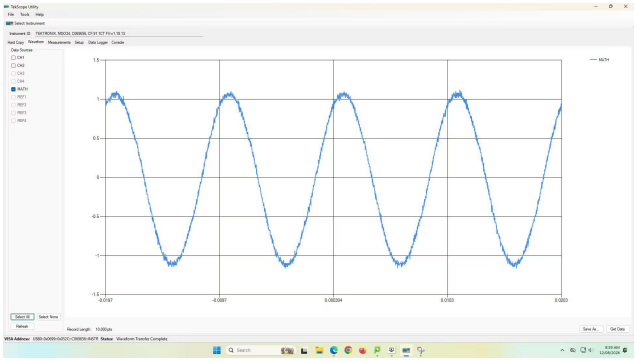


Figure 3. Oscilloscope MATH waveform for CH1 + CH2.

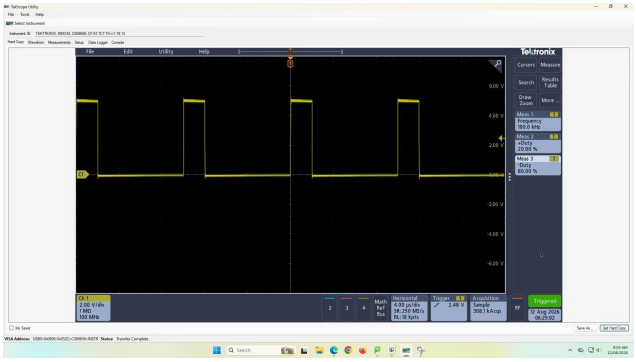


Figure 4. Representative 100 kHz PWM waveform at 20% positive duty.

Nominal positive duty	Frequency	Positive duty	Low-state duty	Error (pp)
20%	100.002 kHz	19.997%	80.003%	-0.003
40%	100.000 kHz	39.998%	60.002%	-0.002
60%	100.001 kHz	59.998%	40.002%	-0.002

All measured PWM frequencies were within 0.002 kHz of 100 kHz, and positive-duty error did not exceed 0.003 percentage point.

4. DACI/LVDAC-EMS DC results

$R_1 = 82\ \Omega$ and $R_2 = 40\ \Omega$ were connected in series. I_1 measured series current, E_1 measured the voltage across R_1, and P_1 was recorded by LVDAC-EMS.

Record	E_1 (V)	I_1 (A)	P_1 (W)
1	13.520	0.163	2.209
2	12.120	0.146	1.768
3	10.800	0.129	1.398
4	9.497	0.113	1.078
5	8.114	0.097	0.787
6	6.708	0.081	0.540

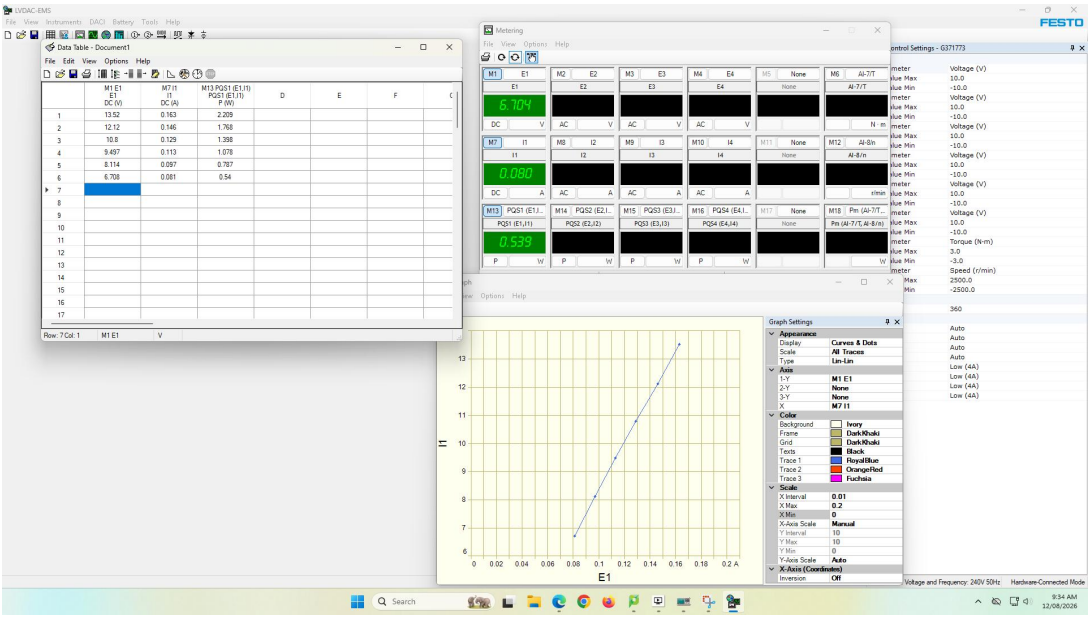


Figure 5. Original LVDAC-EMS data table and E_1-I_1 graph.

4.1 Resistance from the E_1-I_1 slope

$$R_{\text{endpoint}} = \frac{13.520 - 6.708}{0.163 - 0.081} = \frac{6.812}{0.082} = 83.07\ \Omega$$

$$E_1 = (82.67\ \Omega) I_1 + 0.083\ \text{V}, \quad R^2 = 0.9995$$

The endpoint estimate is 1.31% above 82 Ω; the six-point fitted slope is 0.81% above nominal. The regression is preferred because it uses all measurements and is less sensitive to display rounding.

Check: each listed P_1 value is consistent with $E_1 I_1$ within rounding (for example, 13.520 × 0.163 = 2.204 W versus 2.209 W recorded).

5. AC differential-voltage results

The generator was set to 50 Hz and 20 Vpp, but the oscilloscope measured the actual input as 14.600 Vpp. The screenshot identifies CH1 = VBD = 14.600 Vpp and CH2 = VCD = 4.800 Vpp; therefore VBC = VBD - VCD = CH1 - CH2.

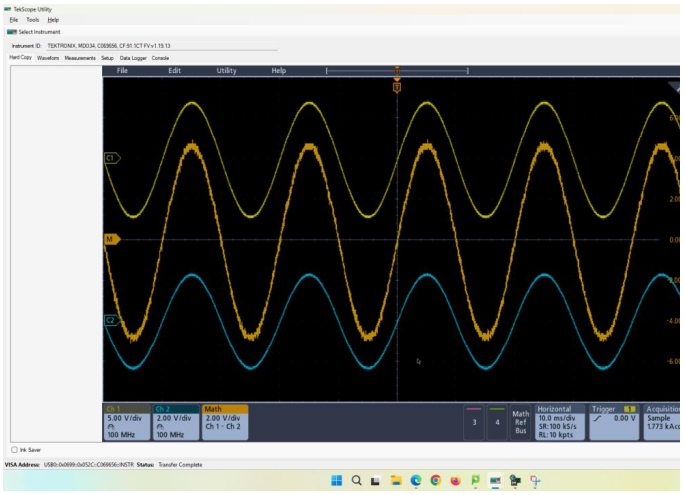


Figure 6. CH1, CH2 and MATH = CH1 - CH2 waveforms.



Figure 7. Lab-Volt readings: E_1 = 3.320 V RMS and I_1 = 0.040 A RMS.

$$V_{\text{in,RMS}} = \frac{14.600}{2\sqrt{2}} = 5.162\ \text{V}; \qquad I_{\text{theory,RMS}} = \frac{5.162}{82 + 40} = 0.04231\ \text{A}$$

$$V_{\text{BC,theory,RMS}} = 5.162 \times \frac{82}{122} = 3.469\ \text{V}$$

Quantity	Measured	Reference	Relative difference
Lab-Volt current, RMS	0.0400 A	0.04231 A theory	-5.46%
Scope MATH, direct RMS	3.355 V	3.469 V theory	-3.30%
Lab-Volt E_1, RMS	3.320 V	3.469 V theory	-4.31%
Scope RMS vs Lab-Volt	3.355 V	3.320 V measured	+1.05%

The scope also reported MATH = 10.119 Vpp. Converting this peak-to-peak value gives 10.119/(2√2) = 3.578 V RMS, 7.76% above the Lab-Volt reading. The direct scope RMS value (3.355 V) is the more valid like-for-like comparison because peak-to-peak detection is more sensitive to noise and waveform distortion.

Using the nominal 20 Vpp setting would predict 7.071 V input RMS, 0.0580 A and 4.754 V across R_1; these describe the set point, not the measured waveform, so they are not used as the primary error reference.

6. Discussion and analysis

Function-generator performance: CH1 amplitude and frequency errors were +2.00% and +0.040%, respectively. The measured phase of 89.892° confirms the intended +90° relation. Phasor addition predicts 1.118 V peak at +26.565° relative to CH1, consistent with the exported MATH trace. PWM frequency and duty measurements were also effectively equal to their programmed values.

DC divider: the E_1-I_1 data were highly linear ($R^2 = 0.9995$). The fitted 82.67 Ω resistance differs from the nominal 82 Ω by only +0.81%, which is compatible with resistor tolerance, lead resistance, instrument resolution and rounding.

AC divider: the direct oscilloscope RMS and Lab-Volt RMS results differ by only 1.05%. Their 3.30%-5.46% differences from theory are mainly explained by the measured 14.600 Vpp input being lower than the nominal 20 Vpp setting, together with component tolerance, source loading, measurement bandwidth and waveform distortion.

Differential direction: both probe grounds were connected to D. Because CH1 measured VBD and CH2 measured VCD, CH1 - CH2 produced VBC. Reversing the MATH order would invert waveform polarity by 180° but would not change RMS or peak-to-peak magnitude.

6.1 Limitations

Only displayed instrument values were available, so resolution and rounding limit the precision of recalculation. The MATH waveform export supports amplitude and frequency but does not independently establish its DC mean. Resistor tolerance, generator output impedance and probe/instrument loading were not measured separately.

7. Conclusion

The laboratory met its core aims. The function generator and oscilloscope produced and verified the required sine, phase, MATH and PWM waveforms. DACI/LVDAC-EMS data gave a fitted resistance of 82.67 Ω ($R^2 = 0.9995$), close to the nominal 82 Ω. For the AC divider, the oscilloscope and Lab-Volt differential-voltage readings agreed within 1.05% on an RMS basis. Reliable interpretation required the actual measured input amplitude and the channel mapping VBC = CH1 - CH2.